

NETWORK ADDRESSING & SUBNETTING PRACTICE

TASK-1 (Network Addressing)

Do the calculations and find the appropriate values in *binary*, *decimal* or *hexadecimal* forms.

a. 89.232.34.118

- Binary |01011001|11101000|00100010|01110110|
- Hexadecimal |59|E8|22|76|

b. 202.9.36.186

- Binary |11001010|00001001|00100100|10111010|
- Hexadecimal |CA|09|24|BA|

c. A9.36.05.DF

- Binary |10101001|00110110|00000101|11011111|
- Decimal |169|54|5|223|

d. 28.5B.29.87

- Binary |00101000|01011011|00101001|10000111|
- Decimal |40|91|41|135|

e. 01101100.11001011.00110110.01110110

- Hexadecimal |6C|CB|36|76|
- Decimal |108|203|54|118|

f. 10011011.00010011.11100010.00111011

- Hexadecimal |9B|13|E2|3B|
- Decimal |155|19|226|59|

TASK-2 (Network Subnetting)

a. Do the calculations and find the appropriate values.

Details and Requirements

- Network Address : 192.168.100.0
- Default Subnet Mask : 255.255.255.0
- Subnets Required : 4 subnets

How many **host bits** do we need to borrow?

- 2

How many **host addresses** per subnet?

- $2^6=64$
- $64-2=62$

What are the **valid subnets**?

- 192.168.100.0, 192.168.100.64, 192.168.100.128, 192.168.100.192

New **Subnet Mask**?

- 11111111.11111111.11111111.11000000
- 255.255.255.192, /26

Subnet	#1	#2	#3	#4
Network Addr.	192.168.100.0	192.168.100.64	192.168.100.128	192.168.100.192
First Host IP	192.168.100.1	192.168.100.65	192.168.100.129	192.168.100.193
Last Host IP	192.168.100.62	192.168.100.126	192.168.100.190	192.168.100.254
Broadcast Addr.	192.168.100.63	192.168.100.127	192.168.100.191	192.168.100.255

TASK-2 (Network Subnetting)

b. Do the calculations and find the appropriate values.

Details and Requirements

- Network Address : 192.168.111.0
- Default Subnet Mask : 255.255.255.0
- Subnets Required : 8 subnets (1-128, 3-32, 4-8 IPs)

How many **host bits** do we need to borrow?

- 1, 3, 5

How many **host addresses** per subnet?

- 126, 30, 6

What are the **valid subnets**?

- 192.168.111.0/25 | 192.168.111.128/27 | 192.168.111.160/27 | 192.168.111.192/27
- 192.168.111.224/29 | 192.168.111.232/29 | 192.168.111.240/29 | 192.168.111.248/29

New **Subnet Masks**?

- 11111111.11111111.11111111.10000000, 255.255.255.128, /25
- 11111111.11111111.11111111.11100000, 255.255.255.224, /27
- 11111111.11111111.11111111.11110000, 255.255.255.248, /29

Subnet	#1	#2	#3	#4
Network Addr.	192.168.111.0	192.168.111.128	192.168.111.160	192.168.111.192
First Host IP	192.168.111.1	192.168.111.129	192.168.111.161	192.168.111.193
Last Host IP	192.168.111.126	192.168.111.158	192.168.111.190	192.168.111.222
Broadcast Addr.	192.168.111.127	192.168.111.159	192.168.111.191	192.168.111.223
Subnet	#5	#6	#7	#8
Network Addr.	192.168.111.224	192.168.111.232	192.168.111.240	192.168.111.248
First Host IP	192.168.111.225	192.168.111.233	192.168.111.241	192.168.111.249
Last Host IP	192.168.111.230	192.168.111.238	192.168.111.246	192.168.111.254
Broadcast Addr.	192.168.111.231	192.168.111.239	192.168.111.247	192.168.111.255